

Computer Science

Middletown High School | Grade 12, Fall Semester 2026

Teacher: Ms. K. Oduya
Room: Lab B
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Course Description

An introduction to programming and computer systems in Python. Students write working programs, reason about how long they take to run, and store data in a database.

Course Units

Weeks 1 to 4	Unit 1: Programming Fundamentals in Python Variables, types, conditionals, loops, functions and reading error messages.
Weeks 5 to 7	Unit 2: Data Structures Lists, stacks which are last in first out, queues which are first in first out, and dictionaries which store key and value pairs. Choose the structure by how data is added and removed.
Weeks 8 to 10	Unit 3: Algorithms, Searching and Sorting Linear search checks each item and is $O(n)$. Binary search halves a sorted list each step and is $O(\log n)$. Bubble sort is $O(n^2)$ and merge sort is $O(n \log n)$.
Weeks 11 to 13	Unit 4: Computer Networks Packets, IP addressing, the client and server model, and basic web requests.
Weeks 14 to 16	Unit 5: Databases and SQL Tables, keys, and SELECT, WHERE and JOIN queries.

Key Dates This Semester

Sep 23	Python Mini-Project due
Oct 19	Mid-Term Exam, covers Units 1 to 3
Dec 4	Final Project due
Dec 15	Final Exam

School wide dates

Sep 8	First day of classes
Oct 19 to Oct 23	Mid-term examination week
Nov 6	Parent teacher conferences
Nov 25 to Nov 27	Thanksgiving break, no classes
Dec 14 to Dec 18	Final examination week

Materials

School laptop or personal machine with Python 3 installed, USB drive or cloud folder for backups.

Grading Policy

25 percent	Programming assignments
30 percent	Final project
10 percent	Quizzes
15 percent	Unit tests
20 percent	Final exam

A = 90 to 100 percent, B = 80 to 89, C = 70 to 79, D = 60 to 69, F = 59 and below.

Late and Make-Up Work

Work handed in late without an excused absence receives a maximum of 50 percent credit. With an excused absence, work is due within one week of returning. It is the student's responsibility to ask what was missed. Make-up tests are held on Thursdays after school and must be booked at least one day ahead.

Academic Honesty

All work submitted must be your own. Copying another student's work, or submitting text or code generated by someone or something else as your own, is treated as academic dishonesty and results in a zero for that assignment and a call home.

Extra Help

Extra help is available Tuesdays and Thursdays, 3:15 to 4:00 pm, in the room listed above.